# Defect-class outcome matrix (RQ2)

Every injected defect is a compile error in safe Rust; in C++ each compiles and is caught, if at all, only at runtime

| Defect class | C++ · ASan | C++ · TSan | Safe Rust (rustc) |
|---|---|---|---|
| **Use-after-free**<br>stale ptr into context map | **detected**<br>at runtime | **detected**<br>at runtime | **prevented**<br>compile-time · E0499 |
| **Double-free**<br>context released twice | **detected**<br>at runtime | **not detected**<br>(clean) | **prevented**<br>compile-time · E0382 |
| **Iterator invalidation**<br>mutate DRB list while iterating | **detected**<br>at runtime | **detected**<br>at runtime | **prevented**<br>compile-time · E0502 |
| **Data race**<br>unsynchronised field update | **not detected**<br>(clean) | **detected**<br>at runtime | **prevented**<br>compile-time · E0373 |
| **Unguarded shared mutation**<br>ctx shared & mutated, no lock | **not detected**<br>(clean) | **detected**<br>at runtime | **prevented**<br>compile-time · E0502 |

prevented at compile time (safe Rust) · compiles in C++, caught at runtime · sanitiser does not detect it